Minseong Cho

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Softlab, Ajou University
Suwon, South Korea
Portfolio
Google Scholar

Research Interests

Active matter dynamics · Elastic Leidenfrost effect · Mechanics of Moisture-Driven Hair Frizz

Education

- **M.S. in Mechanical Engineering** Sep 2025 – Present
  Ajou University, Suwon, South Korea
  - **Advisor:** Prof. Jonghyun Ha
- **B.S. in Mechanical Engineering** Mar 2020 – Aug 2025
  Ajou University, Suwon, South Korea

Research Experience

- **Graduate Research Student** – Softlab, Ajou University Sep 2025 – Present
  - **Mechanics of Moisture-Driven Hair Frizz**
    Keywords: Hygroscopic swelling, Hair frizz, Image processing, Simulation
    Advisors: Prof. Jonghyun Ha, Dr. Yeonsu Jung
  - **Boundary-Driven Active Matter in Soft Hydrogels**
    Keywords: Active matter, Elastic Leidenfrost effect, Image processing
    Advisors: Prof. Jonghyun Ha, Dr. Yeonsu Jung
- **Undergraduate Research Student** – Softlab, Ajou University Mar 2023 – Aug 2025
  - **Boundary-Driven Active Matter in Soft Hydrogels**
    Keywords: Active matter, Elastic Leidenfrost effect, Image processing
    Advisors: Prof. Jonghyun Ha, Dr. Yeonsu Jung

Conferences and Presentations

- **APS Division of Fluid Dynamics Meeting** (Houston, TX, USA) Nov 2025
  Leidenfrost-induced active matter dynamics of spherical hydrogels
- **Asian Congress of Fluid Mechanics** (Seoul, South Korea) Sep 2025
  Leidenfrost-induced active matter dynamics of spherical hydrogels
- **KSME Annual Conference** (Jeju, South Korea) Nov 2024
  Explosive motion of spherical hydrogel via the Leidenfrost effect

Awards & Honors

- **Dean's List** – Ajou University — Spring 2024, Spring 2025
  Recognized for outstanding academic performance (top 5% of the department).
- **Capstone Design Competition** – Ajou University — June 2025
  2nd Place · KRW 1,000,000 prize
  Project: Compliant-mechanism-based knee assistive device for degenerative osteoarthritis prevention.
- **Gyeonggi-do Assistive Device Idea Competition** — Nov 2025
  3rd Place · KRW 300,000 prize
  Awarded by: Gyeonggi Province Rehabilitation Engineering Service Support Center (South Korea)
  Project: Compliant-mechanism-based knee assistive device for older adults.
- **APS DFD Travel/Enabling Grant** — Apr 2026
  US$750 travel cost reimbursement
  Awarded by: APS Division of Fluid Dynamics, American Physical Society
  Purpose: Travel support for participation in the 2025 APS DFD Annual Meeting.
- **Gallery of Fluid Motion** — Apr 2026
  Haecheon Choi GFM Award · KRW 400,000 prize
  Awarded by: The Korean Society of Mechanical Engineers, Fluid Engineering Division
  Title: Explosive motion of spherical hydrogel via the Leidenfrost effect
  Authors: **Minseong Cho**, Yeonsu Jung, and Jonghyun Ha

Publications

- **Mechanics of moisture-driven hair frizz**
  Minseong Cho, Jaehyuk Cho, Hyunna Kim, Yeonsu Jung, and Jonghyun Ha
  ⌛ In preparation
- **Poroelastic swelling time as a quantitative marker of hair damage and treatment**
  Jaehyuk Cho, **Minseong Cho**, Selin Cho, Haesoo Park, Hyun-Na Kim, and Jonghyun Ha
  ⌛ Submitted
- **Boundary-driven activity of a Leidenfrost hydrogel**
  Minseong Cho, Yeonsu Jung, and Jonghyun Ha
  ⌛ Under review

References

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